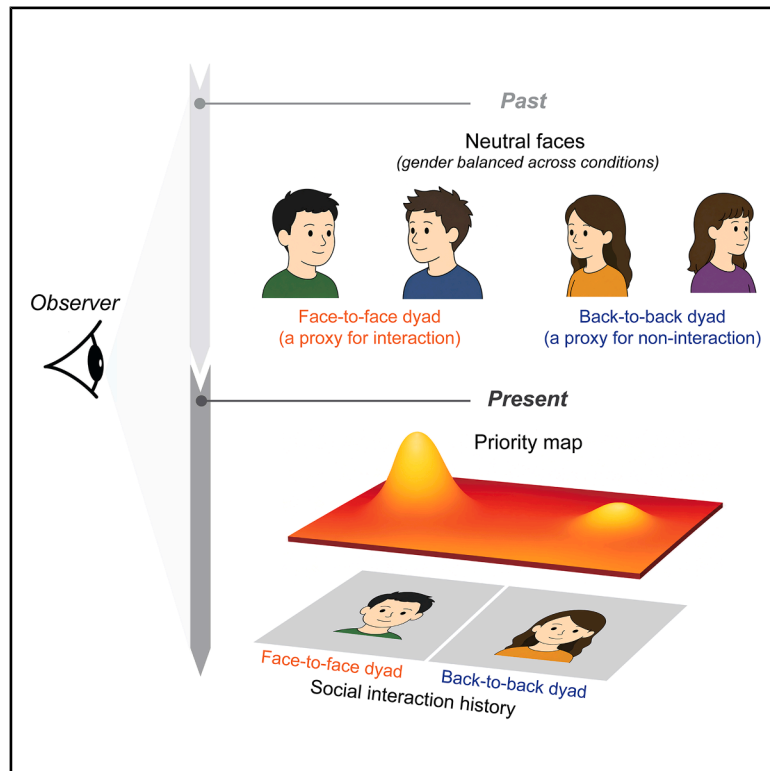


Associative learning of social interaction alters attention to human faces

Graphical abstract



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In brief

Social sciences; Psychology

Highlights

- Social interaction history shapes subsequent attentional selection
- Faces previously seen interacting capture attention more than non-interacting ones
- Attentional bias disappears with inverted faces and nonsocial directional stimuli
- These results extend history-driven attention to abstract social structures



Article

Associative learning of social interaction alters attention to human faces

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SUMMARY

Beyond current goals and physical salience, selection history reflects how past experience automatically biases visual attention. While existing frameworks characterize how we learn to prioritize non-social regularities such as reward and spatial probability, whether selection history can incorporate abstract social structures governing interactions between others remains unknown. Across four experiments ($N = 144$), we examined whether observed social configurations produce lasting attentional biases toward individual agents. Participants first viewed pairs of human faces arranged in interacting (face-to-face) or non-interacting (back-to-back) configurations. In a subsequent cueing task, faces previously shown interacting captured attention more strongly than those shown non-interacting. This effect was abolished by face inversion and absent for non-social directional stimuli (arrows, animal faces), ruling out contributions from low-level visual features or general directional cueing. Together, these findings indicate that social interaction history induces lasting changes in the priority map, endowing faces with learned attributes that guide selection in social environments.

INTRODUCTION

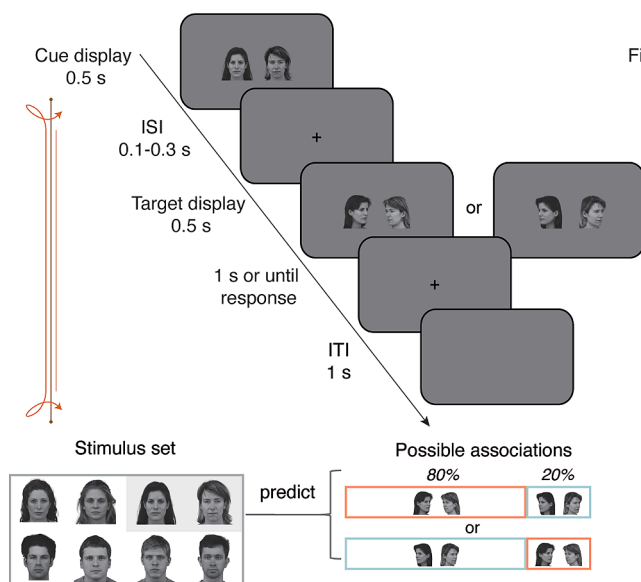
Attentional priority is traditionally viewed as a dichotomy between top-down goals and bottom-up salience. A third construct, selection history, has recently been proposed to account for how past experience can automatically bias attentional selection in the present.^{1–4} This influence of selection history is commonly instantiated through associative learning, whereby statistical regularities such as reward contingencies and spatial probabilities of targets or distractors, are encoded and used to guide attentional selection.^{5–7} However, the scope of selection history has largely been confined to non-social physical regularities. A fundamental yet unresolved question is whether selection history can incorporate higher-order social regularities into attentional priority maps, thereby extending experience-driven attention from non-social to social domains.

While individual faces represent the most significant entities in our social world, interactions between them arguably constitute the most pervasive regularities. In everyday life, we are repeatedly exposed to others' interactions from a third-person perspective, often referred to as third-party encounters or people-watching.^{8–10} A growing body of research demonstrates that observers can rapidly extract socially relevant information from processing such interactions, including forming impressions and identifying potential partners.¹¹ Although these infer-

ences were traditionally attributed to higher-level cognitive processes, recent theories propose that social interactions rely primarily on visual computations.¹² Consistent with this view, interacting configurations (face-to-face dyads) are recognized more accurately^{13,14} and detected more rapidly^{15–18} than non-interacting configurations (back-to-back dyads), suggesting prioritized processing of dyadic interactions. However, these findings focused on how we respond to currently perceived social interactions. It remains unknown whether previously observed social interactions leave a lasting “history” that biases attention toward the individuals involved. A prior study hinted at this possibility by showing that direct experience of joint gaze with a particular face modulates subsequent gaze-cued attention toward the same face.¹⁹ Yet whether merely observing third-party social interactions can leave a lasting trace that biases attention toward the individuals involved remains unknown. Here, we asked whether faces frequently embedded in such interactions become internalized as learned attributes of face representations to guide attention in future encounters. Addressing this question is critical for understanding how associative learning of abstract social relationships alters attentional selection. Moreover, establishing a link between social learning and attentional biases may shed new light on the mechanisms underlying atypical face-related attention in psychiatric conditions such as autism and social anxiety.^{20–22}



A Training task



B Test task

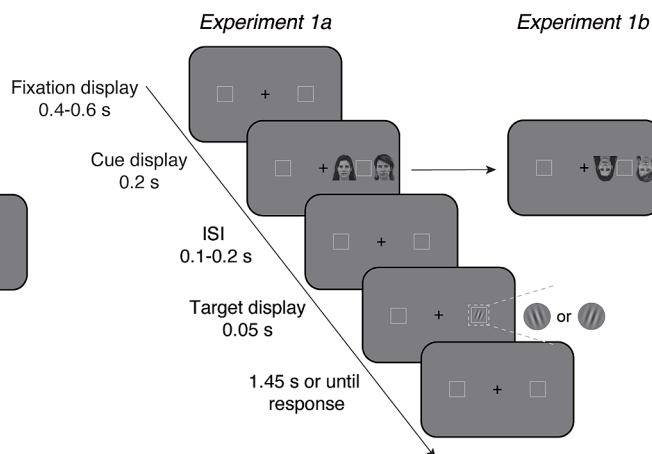


Figure 1. Tasks for Experiment 1a and 1b

(A) Trial sequence and stimuli used in the training phase. Participants reported whether the target dyad was face-to-face or back-to-back. The associations for each face pair were predetermined (see inset learned association) and randomized across participants. The full stimulus set of eight faces is shown at the lower left, along with an example of the pre-specified pairings.

(B) Trial sequence of the cued orientation discrimination task during the test phase. Participants reported the orientation of the target Gabor (clockwise or counterclockwise) following the face cue. The target appeared at the cued location in 50% of trials. The dashed lines around Gabor patches are for illustrative purposes and were not shown in the actual experiment.

To examine whether the history of social interaction influences attention to faces, we first trained participants in a task where specific face pairs (with direct gaze toward the viewer) had a high probability of appearing in either interacting (face-to-face dyads; 80% probability) or non-interacting (back-to-back dyads; 80% probability) configurations. The face-to-face dyads have been consistently employed as a proxy for social interaction in the literature.^{23–27} In a subsequent test task, these trained faces were presented as task-irrelevant spatial cues to measure attentional capture. We found that face pairs previously embedded in interacting configurations produced stronger attentional capture, an effect that disappeared when the faces were inverted. This effect persisted even when a single face from a learned dyad was presented in isolation. In contrast, the effect did not generalize to non-social directional stimuli (arrows) or non-human faces (animal faces). Our findings provide the first evidence that observing social interactions can rapidly and automatically bias attention toward human faces, highlighting social learning as a key component of experience-driven attention.

RESULTS

Social interaction history enhances attentional capture by face dyads

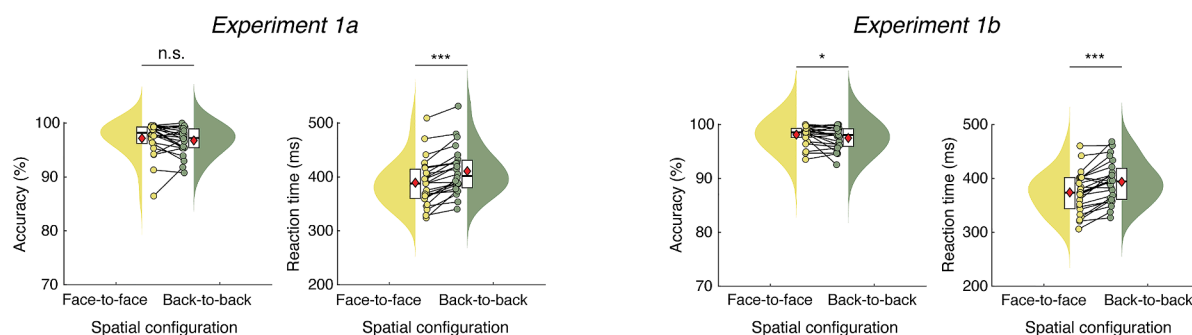
In Experiment 1a, we investigated whether attention can be selectively tuned to faces previously embedded in interacting configurations. Specifically, we first used an associative learning

task to pair faces in either interacting (face-to-face) or non-interacting (back-to-back) dyads (Figure 1A). For instance, two specific face pairs with direct gaze (one female pair and one male pair) were assigned an 80% probability of appearing in interacting configurations and a 20% probability in non-interacting configurations. Then, in the test phase, we used a cued discrimination task where these face pairs served as uninformative spatial cues for a subsequent target, allowing us to measure their degree of attentional capture (Figure 1B). To further determine whether this effect is contingent upon holistic face processing rather than low-level facial features, Experiment 1b replicated the test phase using inverted face pairs (Figure 1B, right panel).

In the training task of Experiment 1a, participants responded significantly faster to face-to-face targets than back-to-back targets (389 ms vs. 411 ms; $t(23) = -7.23$, $p < 0.001$, Cohen's $d = -1.48$, 95% CI = $[-2.05, -0.89]$; Figure 2A, left panel), replicating previous findings and confirming our stimulus arrangements.^{15,16} No significant difference in accuracy was observed (97.2% vs. 96.8%; $t(23) = 1.00$, $p = 0.326$, Cohen's $d = 0.21$, 95% CI = $[-0.20, 0.61]$).

In the test task of Experiment 1a, previously learned social interaction enhanced the attentional cueing effect (Figure 2B, left panel). A two-way repeated-measures analysis of variance (ANOVA) (learned configuration: face-to-face vs. back-to-back $\times$ cue validity: valid vs. invalid) on discrimination accuracy revealed a significant main effect of cue validity ($F(1,23) = 28.06$, $p < 0.001$, $\eta_p^2 = 0.55$). Crucially, there was a significant

A Training performance



B Test performance

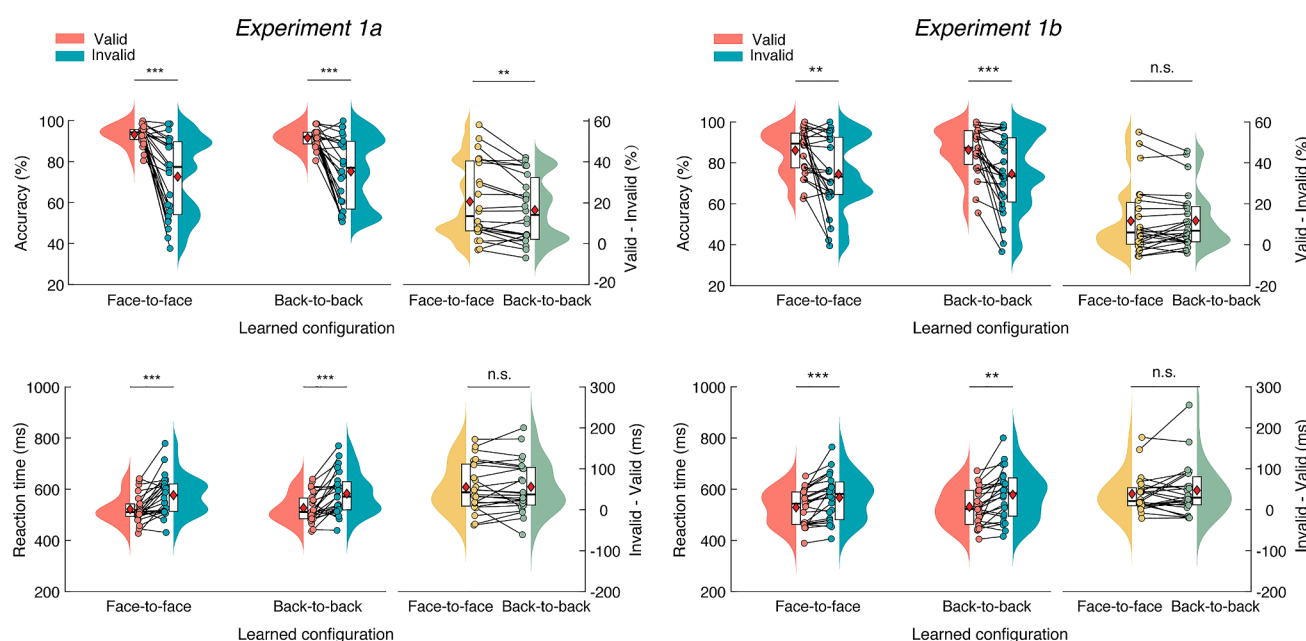


Figure 2. Results for Experiment 1a and 1b

(A) Accuracy and RT during the training task in Experiment 1a (left panel) and 1b (right panel). Each subplot shows the distribution of accuracy or RT across the face-to-face and back-to-back target conditions. The boxplots show the interquartile range, the horizontal black line represents the median, and the red dot indicates the mean for each condition. The lines connecting the distributions represent individual performance across conditions.

(B) Accuracy and RT during the test task in Experiment 1a (left panel) and 1b (right panel). Each subplot shows the accuracy or RT for each cue condition (left Y axis), along with the cueing effect indices (valid–invalid condition; shown on the right Y axis of each subplot). *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. n.s. = not significant.

interaction effect ($F(1,23) = 11.26$, $p = 0.003$, $\eta_p^2 = 0.33$), indicating stronger attentional capture for face pairs previously embedded in interacting compared to non-interacting configurations. Follow-up comparisons showed that accuracy was numerically higher for faces previously embedded in face-to-face than back-to-back configurations on valid trials ($t(23) = 1.83$, $p = 0.080$, Cohen's $d = 0.37$, 95% CI = $[-0.05, 0.78]$), and significantly lower on invalid trials ($t(23) = -2.52$, $p = 0.019$, Cohen's $d = -0.51$, 95% CI = $[-0.94, -0.08]$). No main effect of learned configuration was found ($F(1,23) = 0.53$, $p = 0.473$, $\eta_p^2 = 0.02$). The same ANOVA on reaction times (RTs) showed only a significant main effect of cue validity ($F(1,23) = 20.57$,

$p < 0.001$, $\eta_p^2 = 0.47$), with no other significant effects ($p > 0.084$), ruling out the possibility of speed-accuracy trade-offs (e.g., improved accuracy due to slowed responses).

In contrast, the attentional bias toward faces previously embedded in interacting configurations disappeared when inverted face pairs were used during the test phase in Experiment 1b. During training, participants showed similar advantages in identifying face-to-face versus back-to-back targets (RT: 374 ms vs. 394 ms: $t(23) = -5.56$, $p < 0.001$, Cohen's $d = -1.14$, 95% CI = $[-1.64, -0.61]$; accuracy: 98.1% vs. 97.5%: $t(23) = 2.21$, $p = 0.038$, Cohen's $d = 0.45$, 95% CI = $[0.03, 0.87]$; Figure 2A, right panel). In the test task, a two-way

repeated-measures ANOVA (learned configuration $\times$ cue validity) on discrimination accuracy revealed only a significant main effect of cue validity ($F(1,23) = 13.91$, $p = 0.001$, $\eta_p^2 = 0.38$; Figure 2B, right panel). No significant main effect of learned configuration ($F(1,23) = 0.03$, $p = 0.872$, $\eta_p^2 = 0.001$) or interaction ($F(1,23) = 0.02$, $p = 0.891$, $\eta_p^2 < 0.001$) was observed. The same ANOVA on RTs revealed only a significant main effect of cue validity ($F(1,23) = 16.56$, $p < 0.001$, $\eta_p^2 = 0.42$), with no other significant effects ($ps > 0.082$). To further evaluate the strength of evidence for the lack of an interaction effect, we conducted parallel Bayesian analyses²⁸ by comparing a null model containing two main effects and subject-related effects to an interaction model that included all terms of the null model plus the interaction. Bayesian analyses provided moderate evidence for the null model on accuracy ($BF_{01} = 3.46$) and anecdotal evidence for the null model on RT ($BF_{01} = 2.12$). To validate whether the effect of learned configuration on attentional cueing differed significantly between Experiments 1a and 1b, we used a three-way mixed ANOVA (learned configuration $\times$ cue validity $\times$ experiment) on accuracy and found a significant three-way interaction ($F(1,46) = 4.87$, $p = 0.032$, $\eta_p^2 = 0.10$).

Our findings from Experiment 1 demonstrate that social interaction history enhanced attentional capture by faces previously embedded in interacting configurations, even when these faces now appear in non-interactive, task-irrelevant contexts. Importantly, this bias was present only for upright faces, not inverted ones, indicating that social interaction history modulated holistic face processing rather than low-level facial features.

Attentional bias for interacting faces is not driven by directional “hot-spots”

An alternative account for the enhanced attentional capture by social interaction history is the attentional “hot-spots” effect, wherein directional information from two items converges to bias attention toward their center.^{16,29} To determine whether our findings reflect such a domain-general or a specialized processing of social entities, Experiment 2 substituted faces with non-social arrows. By arranging these arrows in point-to-point versus base-to-base configurations (Figures 3A and 3B), we created directional cues equivalent to the face dyads but devoid of social meaning, thus testing whether the observed bias is contingent upon social interaction or merely directional geometry.

We conducted the same analyses as in Experiment 1. During the associative learning task, RTs were significantly faster for point-to-point than base-to-base arrow pairs (390 ms vs. 405 ms: $t(23) = -3.54$, $p = 0.002$, Cohen's $d = -0.72$, 95% CI = $[-1.17, -0.27]$; Figure 3C), similar to the results in Experiment 1 and replicated previous reports of attentional advantages for non-social stimuli with directional cues.^{16,17} No significant difference in accuracy was observed between the two configurations (96.9% vs. 96.8%: $t(23) = 0.07$, $p = 0.944$, Cohen's $d = 0.02$, 95% CI = $[-0.39, 0.41]$).

Despite the facilitated responses to point-to-point arrows during training, test performance showed no effect of the learned configurations (Figure 3D). A two-way repeated-measures ANOVA (learned configuration: point-to-point vs. base-to-base $\times$ cue validity: valid vs. invalid) on discrimination accuracy

revealed only a main effect of cue validity ($F(1,23) = 20.40$, $p < 0.001$, $\eta_p^2 = 0.47$), with no main effect of learned configuration ($F(1,23) = 0.59$, $p = 0.449$, $\eta_p^2 = 0.03$) or interaction ($F(1,23) = 0.11$, $p = 0.738$, $\eta_p^2 < 0.01$). Bayesian analyses provided moderate evidence for the absence of an interaction effect on accuracy ($BF_{01} = 3.44$). The same analysis applied to RTs also showed a main effect of cue validity ($F(1,23) = 17.15$, $p < 0.001$, $\eta_p^2 = 0.43$), but no other significant effects ($ps > 0.338$), and anecdotal Bayesian evidence for the absence of an interaction ($BF_{01} = 2.42$). To directly examine whether the effects of learned configurations differed significantly between social (Experiment 1a) and non-social stimuli (Experiment 2), we conducted a three-way mixed ANOVA (learned configuration $\times$ cue validity $\times$ experiment) on discrimination accuracy, which revealed a significant three-way interaction ($F(1,46) = 4.34$, $p = 0.043$, $\eta_p^2 = 0.09$). These results suggest that stimuli previously associated with non-social directional configurations were insufficient to produce attentional biases.

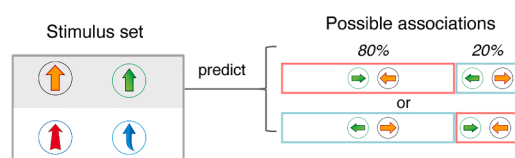
Socially-induced attentional capture is specific to human faces

To examine whether the attentional effects observed in Experiment 1 were specific to human faces or general to facial stimuli across species, we replaced the face stimuli with animal faces (cats) while keeping the experimental design identical in Experiment 3a (Figures 4A and 4B).

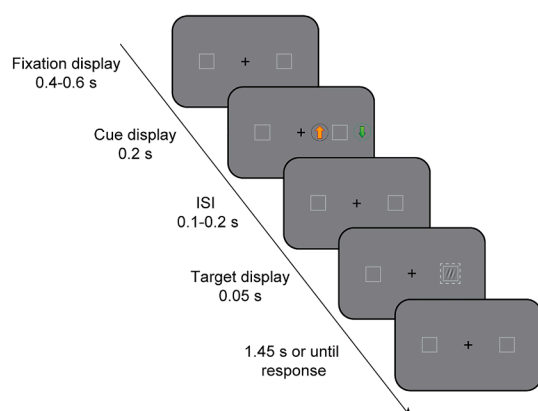
Consistent with the previous experiments, participants responded faster to face-to-face than back-to-back targets during training (377 ms vs. 396 ms: $t(23) = -4.99$, $p < 0.001$, Cohen's $d = -1.02$, 95% CI = $[-1.51, -0.52]$; Figure 4C), with no significant differences in accuracy (97.5% vs. 96.7%: $t(23) = 1.33$, $p = 0.196$, Cohen's $d = 0.27$, 95% CI = $[-0.14, 0.68]$). Test performance was similar to that observed with arrow stimuli in Experiment 2 (Figure 4D). A two-way repeated-measures ANOVA (learned configuration: face-to-face vs. back-to-back $\times$ cue validity: valid vs. invalid) on discrimination accuracy revealed a significant main effect of cue validity ($F(1,23) = 12.24$, $p = 0.002$, $\eta_p^2 = 0.35$), but no main effect of learned configuration ($F(1,23) = 0.45$, $p = 0.508$, $\eta_p^2 = 0.02$) or interaction ($F(1,23) = 0.54$, $p = 0.469$, $\eta_p^2 = 0.02$). The same analysis on RTs showed only a main effect of cue validity ($F(1,23) = 14.90$, $p < 0.001$, $\eta_p^2 = 0.39$), with no other significant effects ($ps > 0.633$). Bayesian analyses supported the lack of an interaction effect for both accuracy ($BF_{01} = 3.20$) and RT ($BF_{01} = 3.57$). To compare effects between Experiments 1a and 3a, we conducted a three-way mixed ANOVA (learned configuration $\times$ cue validity $\times$ experiment) on discrimination accuracy, which revealed a significant three-way interaction ($F(1,46) = 5.03$, $p = 0.030$, $\eta_p^2 = 0.10$). These results suggest that the learned attentional bias from prior social interaction experience did not extend to non-human entities.

Before concluding that the attentional bias was specific to human faces, one possible concern is whether the null effect in Experiment 3a reflected a coarser, category-level processing of cat faces during training. Although Experiments 1a and 3a used the same training protocol, participants may have automatically extracted individual identities for human faces in Experiment 1a, but only category-level information for cat faces in

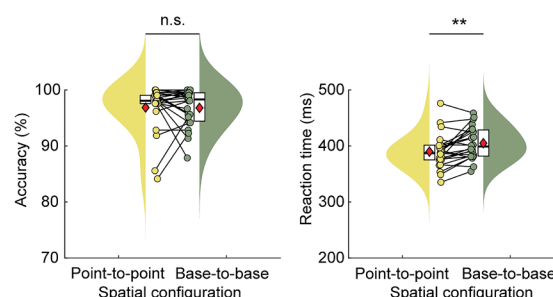
A Training stimuli



B Test task



C Training performance



D Test performance

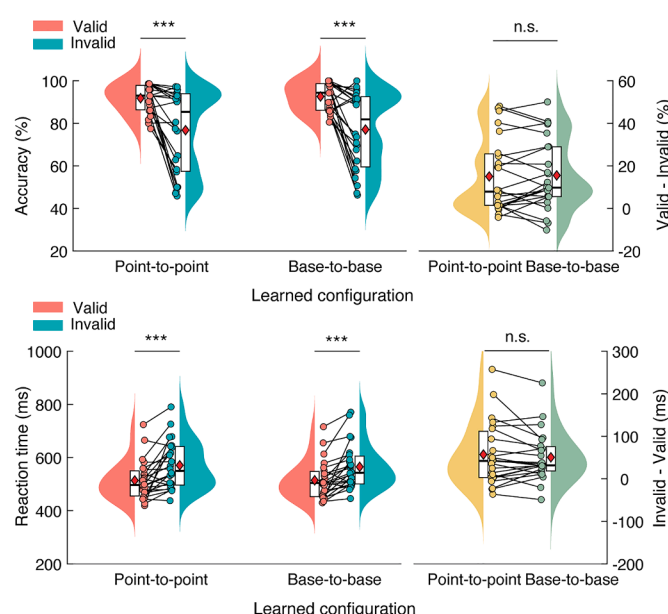


Figure 3. Stimuli, tasks and results for Experiment 2

(A) Stimuli and pre-defined association during the training task. The trial sequence was the same as that in Experiment 1.

(B) Trial sequence of the test task.

(C) Accuracy and RT during the training task.

(D) Accuracy and RT during the test task. Each subplot shows the distribution of accuracy or RT for each condition. The boxplots show the interquartile range, the horizontal black line represents the median, and the red dot indicates the mean for each condition. The lines connecting the distributions represent individual performance across conditions. *** $p < 0.001$, ** $p < 0.01$. n.s. = not significant.

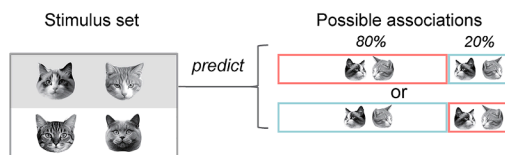
Experiment 3a. To investigate this possibility, we modified the training task such that participants needed to discriminate among the cat faces presented as cue stimuli (Figure 5A), ensuring that they reliably encoded the identity of each cat face.

Consistent with Experiment 3a, participants again showed an advantage for face-to-face over back-to-back targets during training (RT: 445 ms vs. 469 ms: $t(23) = -5.80$, $p < 0.001$, Cohen's $d = -1.18$, 95% CI = $[-1.70, -0.65]$; accuracy: 96.1% vs. 94.1%: $t(23) = 4.32$, $p < 0.001$, Cohen's $d = 0.88$, 95% CI = $[0.40, 1.35]$; Figure 5B). Importantly, performance in the probe trials was also high ($85.61\% \pm 7.53\%$), confirming that participants successfully encoded identities of individual cat faces.

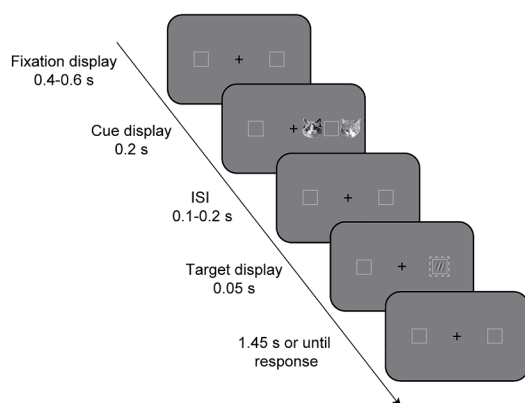
Having established that individual cat faces were processed during training, we next examined whether attentional effects

in the test task differed between face-to-face and back-to-back conditions (Figure 5C). A two-way repeated-measures ANOVA on discrimination accuracy (learned configuration: face-to-face vs. back-to-back $\times$ cue validity: valid vs. invalid) revealed a significant main effect of cue validity ($F(1,23) = 11.85$, $p = 0.002$, $\eta_p^2 = 0.34$), but no effect of learned configuration ($F(1,23) = 0.74$, $p = 0.399$, $\eta_p^2 = 0.03$) and no interaction ($F(1,23) = 0.35$, $p = 0.562$, $\eta_p^2 = 0.02$). The same analysis on RTs showed only a main effect of cue validity ($F(1,23) = 16.16$, $p < 0.001$, $\eta_p^2 = 0.41$), with no other significant effects ($p > 0.527$). Bayesian analyses further supported the absence of an interaction effect for both accuracy ($BF_{01} = 3.06$) and RTs ($BF_{01} = 3.33$). These results indicate that the attentional bias derived from prior social interaction experience did not generalize to non-human entities.

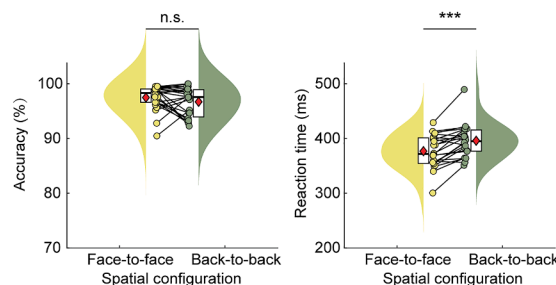
A Training stimuli



B Test task



C Training performance



D Test performance

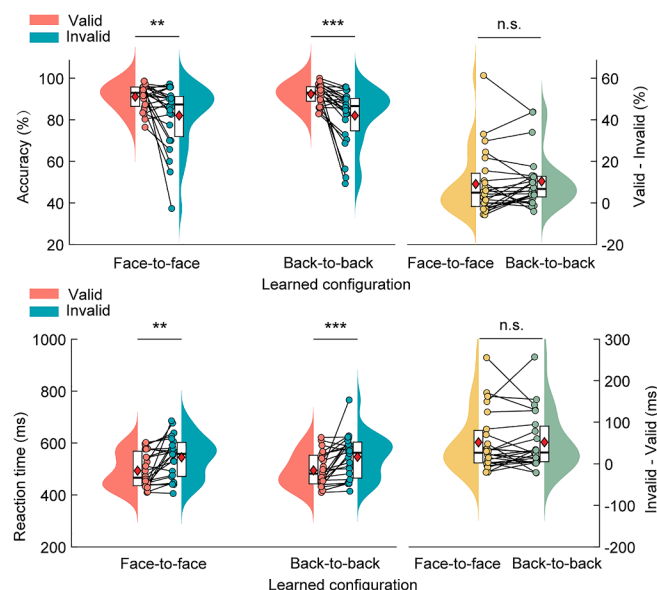


Figure 4. Stimuli, tasks and results for Experiment 3a

(A) Stimuli and pre-defined association during the training task.

(B) Trial sequence of the test task.

(C) Accuracy and RT during the training task.

(D) Accuracy and RT during the test task. Each subplot shows the distribution of accuracy or RT for each condition. The boxplots show the interquartile range, the horizontal black line represents the median, and the red dot indicates the mean for each condition. The lines connecting the distributions represent individual performance across conditions. *** $p < 0.001$, ** $p < 0.01$. n.s. = not significant.

Although our sample size, determined by an *a priori* power analysis, was sufficient to detect learned attentional effects for human faces, one might still wonder whether the null effects observed for other stimuli (arrow, cat faces) reflected insufficient power to detect a smaller effect. To address this concern, we pooled data from the three experiments (Experiments 2, 3a, 3b; $N = 72$) and re-ran the ANOVA, which again showed no significant interaction (learned configuration $\times$ cue validity: $p = 0.326$; $BF_{01} = 3.54$).

Social interaction history modulates attentional priority of individual faces

Prior work suggests that face-to-face dyads are perceptually grouped into holistic units.^{24,30,31} However, it remains unclear how the social grouping during associative learning influences the processing of these individual entities. To address this ques-

tion, we presented single faces in isolation during the test phase to examine whether a history of social interaction facilitates or impairs the processing of individual faces (Figures 6A and 6B).

Replicating the results of Experiment 1, we found faster RTs for face-to-face dyads than back-to-back dyads during training (381 ms vs. 393 ms: $t(23) = -3.94$, $p < 0.001$, Cohen's $d = -0.80$, 95% CI = $[-1.26, -0.34]$; Figure 6C). No significant difference was found in accuracy (96.8% vs. 97.0%; $p = 0.781$). Importantly, test performance was similar to that observed in Experiment 1a (Figure 6D). A two-way repeated-measures ANOVA (learned configuration $\times$ cue validity) on discrimination accuracy revealed significant main effects of cue validity ($F(1,23) = 64.08$, $p < 0.001$, $\eta_p^2 = 0.74$), learned configuration ($F(1,23) = 4.54$, $p = 0.044$, $\eta_p^2 = 0.17$) and an interaction ($F(1,23) = 5.16$, $p = 0.033$, $\eta_p^2 = 0.18$). This interaction effect indicates stronger attentional capture when the cued faces

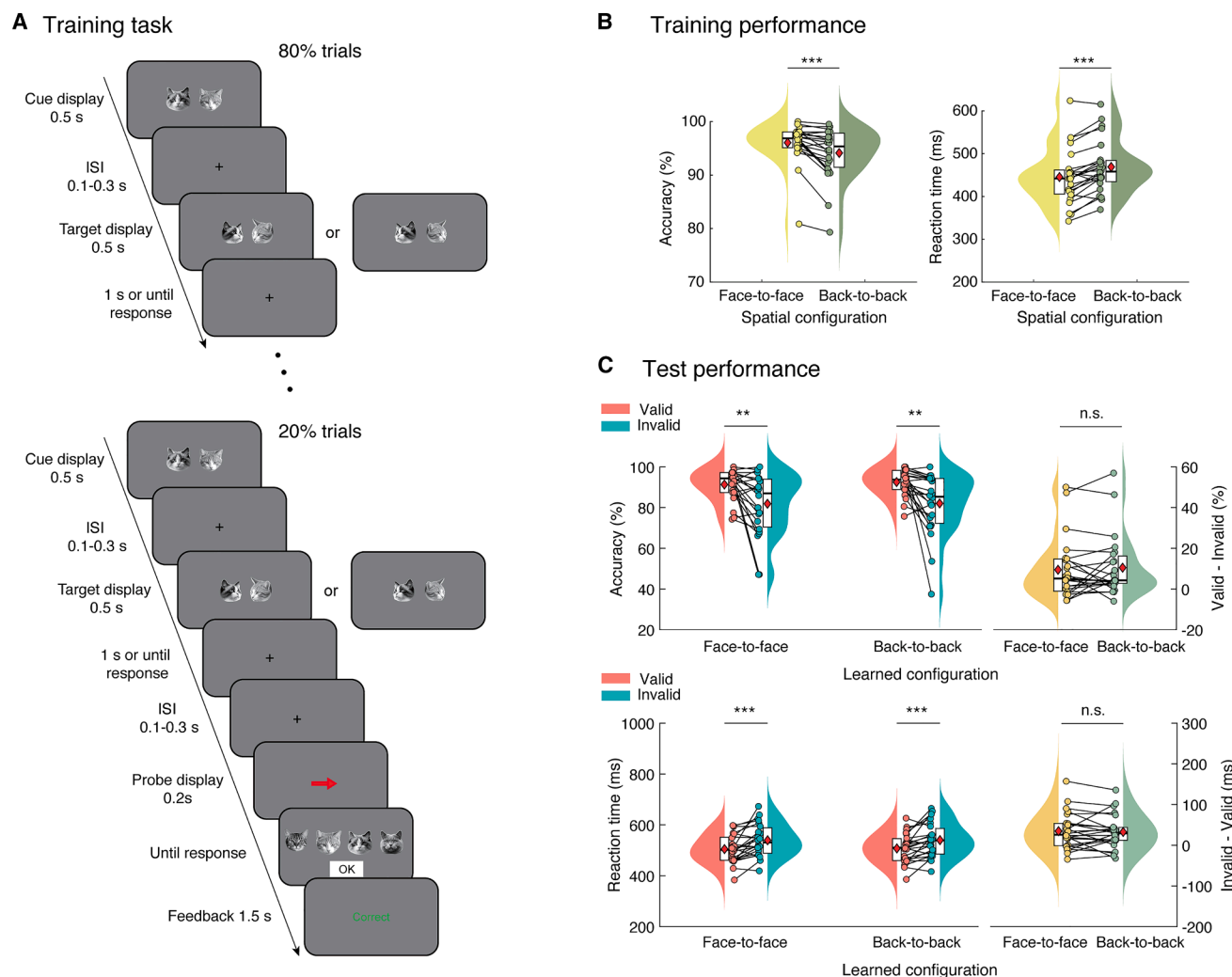


Figure 5. Stimuli, tasks and results for Experiment 3b

(A) Trial sequence of the training task. During training, 20% of trials were probe trials in which participants identified cat faces at a cued location from the preceding display; the remaining 80% of trials followed the standard training task.

(B) Accuracy and RT during the training task.

(C) Accuracy and RT during the test task. Each subplot shows the distribution of accuracy or RT for each condition. The boxplots show the interquartile range, the horizontal black line represents the median, and the red dot indicates the mean for each condition. The lines connecting the distributions represent individual performance across conditions. *** $p < 0.001$, ** $p < 0.01$. n.s. = not significant.

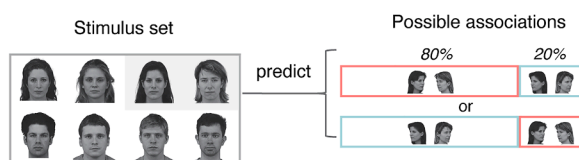
had previously been embedded in interacting configurations compared to non-interacting ones. This interaction remained robust regardless of the distance of the cue from fixation (near vs. far), with no interaction involving distance ($p = 0.83$; $BF_{01} = 3.48$). The same analysis on RTs revealed only a main effect of cue validity ($F(1,23) = 15.71$, $p < 0.001$, $\eta_p^2 = 0.41$), with no other significant effects ($ps > 0.745$). A cross-experiment comparison using a three-way mixed ANOVA (learned configuration $\times$ cue validity $\times$ experiment) on accuracy from Experiments 1a and 4 revealed no significant three-way interaction ($F(1,46) = 0.22$, $p = 0.639$, $\eta_p^2 < 0.01$). Bayesian analysis provided moderate evidence in favor of the null model over the interaction model ($BF_{01} = 8.36$). These results suggest that the attentional effects driven by social interaction history

on learned faces were comparable, whether the faces were presented in their trained dyads or in isolation.

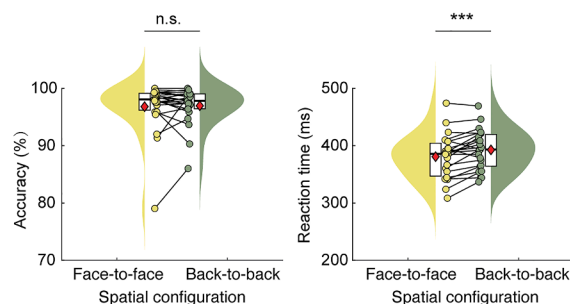
DISCUSSION

Understanding group dynamics requires monitoring interpersonal relationships. The present study shows that the visual system learns from observed social interactions to construct enduring attentional priorities for individual faces involved, extending prior work that merely focused on the prioritization of visual stimuli signaling social interactions.^{13,15,18} Using an associative learning task, we paired face dyads with either interacting (face-to-face) or non-interacting (back-to-back) configurations. Faces previously embedded in interacting

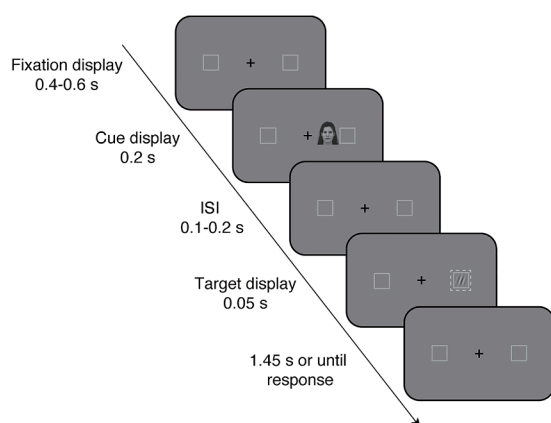
A Training stimuli



C Training performance



B Test task



D Test performance

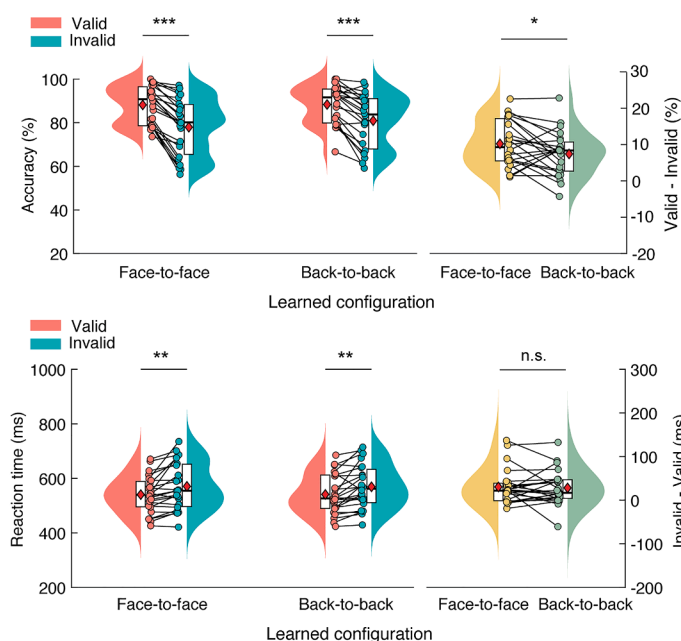


Figure 6. Stimuli, tasks and results for Experiment 4

(A) Stimuli and pre-defined association during the training task.

(B) Trial sequence of the test task.

(C) Accuracy and RT during the training task.

(D) Accuracy and RT during the test task. The boxplots show the interquartile range, the horizontal black line represents the median, and the red dot indicates the mean for each condition. The lines connecting the distributions represent individual performance across conditions. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. n.s. = not significant.

configurations elicited stronger attentional capture than those linked to non-interacting ones. This bias was abolished by face inversion and did not generalize to non-social directional stimuli (arrow or animal faces), ruling out explanations based on low-level visual features and directional cueing.^{16,29} Furthermore, the bias persisted when a single face from the learned dyads was presented, suggesting that social interaction history enhanced individual face representations rather than creating a unitized dyadic chunk. Together, these findings demonstrate that social interaction history can be internalized through associative learning as learned attributes of face representations, endowing particular faces with enhanced attentional priority.

Our findings advance the framework of experience-driven attention by establishing social interaction history as a core determinant of attentional control. While the concept of selection history explains how prior experience shapes attentional priority,²⁻⁴ existing evidence has focused almost exclusively on non-social physical regularities established via associative learning, such as feature-reward associations or spatial probabilities.^{4-7,32,33} Our paradigm follows the same associative learning logic, but with individual faces serving as predictive cues and specific dyadic configurations (i.e., socially meaningful information) serving as the associated outcomes. By repeatedly pairing individual faces with particular configurations at a given probability (e.g., predicting interacting configurations at 80%

probability), participants came to associate certain faces with specific social relationships, as confirmed by faster responses to predicted than unpredicted configurations across stimulus domains ($p < 0.022$). Building on this logic, we show that these learning principles extend to higher-order social structures, connecting attentional selection with broader research on structural learning.^{34,35} Just as the visual system utilizes physical configurations to guide behavior, for instance, utilizing familiarity with object configurations (e.g., a book on a table) to facilitate visual detection³⁶ or constructing cognitive maps of spatial landmarks to support navigation,³⁷ our results demonstrate that social relationships are similarly internalized to guide early selection. These findings point toward a unified computational architecture where the brain recruits common associative learning mechanisms to prioritize both physical regularities and socially meaningful information.

Critically, the prioritized attention observed in our study reflects learned effects that are specific to human faces, rather than a byproduct of directional cueing. Consistent with this notion, recent work has similarly shown that social interactions play a unique role in working memory maintenance relative to nonsocial relations.²⁷ Previous research has frequently relied on face-to-face configurations that generate an attentional “hot-spots” at the center of a dyad,^{15,18,24} driven by salient directional signals such as gaze or body orientation.²⁹ Indeed, similar search advantages have been reported for both social entities (e.g., faces or bodies) and non-social objects (arrows, lamps, cars, and so forth), suggesting that many “social” effects may be confounded by spatial orienting biases.^{16,17} In contrast, our learning-based paradigm dissociated social interaction history from directional cueing, ruling out the low-level directional account. Under these conditions, attentional prioritization emerged as a function of prior social experience, demonstrating that social interaction history itself enhances attentional priority. This explanation was further evidenced by the absence of comparable biases for non-social stimuli (arrows or animal faces). By isolating social interaction history from directional cues, our study reveals a genuine mechanism through which social interactions bias attention. Note that this mechanism differs from previous work demonstrating that prior social interaction modulates the gaze cueing effect.¹⁹ While their findings hinged on active engagement with gaze cues in a second-person paradigm, our study examined third-party social interactions from an observer’s perspective, thereby isolating configural face processing from gaze-following behavior.

We found stronger attentional capture for individuals previously encoded as part of interacting dyads, suggesting that associative learning enhanced individual face representation. Although whether retrieval of features or event elements is facilitated by the presence of other bound elements remains debated,^{38,39} prior studies reveal chunking effects in the social domain: interacting individuals are integrated into larger representational units, improving memory for paired individuals.^{24,30,40} Our results extend these findings by showing facilitated processing of individual faces even in the absence of co-occurring elements, likely due to social grouping during training that strengthened perceptual representations. This interpretation aligns with behavioral evidence that social grouping of

facing dyads automatically influences perception, even when stimuli are rendered invisible,^{13,18} as well as with neuroimaging evidence of enhanced representations in visual cortex for face-to-face than back-to-back dyads.^{41,42}

Building on the view that social learning may involve reward-related mechanisms,⁴³ we propose that reward-based learning offers a parsimonious account for our findings. Social interactions are often inherently rewarding,⁴⁴ supported by neuroimaging evidence of dopaminergic activation during their processing^{45,46} and behavioral findings that face-to-face dyads are rated as more socially valuable than back-to-back dyads.²⁵ Extending prior work showing that neutral stimuli paired with non-social rewards (e.g., money, food^{1,5,6,47–49}) and positive social signals (e.g., faces exhibiting positive expressions^{50–52}) acquire attentional priority, our study shows that individuals can similarly learn to associate faces with interacting scenes that convey social rewards, thereby enhancing the salience of those faces. Future research should use high-resolution neuroimaging^{53,54} to test interactions between the reward system and socially specialized brain regions, particularly areas along the proposed third visual pathway for social perception.^{42,55–57}

Previous studies show that infants can discriminate between facing and non-facing dyads early in development (within the first year). For instance, 9-month-old infants encode objects better when the objects were jointly attended by two people oriented toward each other rather than away,⁵⁸ and 10-month-olds can distinguish between two people in mutual versus averted gaze.⁵⁹ However, a distinct preference for static face-to-face dyads emerges later (around age five), accompanied by increased ratings of social value associated with facingness.²⁵ These findings indicate that accumulated social experience is necessary to establish the social value of human interaction. This aligns with recent theories proposing that associative learning is the fundamental mechanism for encoding causal and structural regularities over development.⁶⁰ Consistent with this notion, our results suggest that the social learning of human interactions shapes attention to individual faces. Furthermore, linking social learning to attentional biases may inform atypical behaviors in autism or social anxiety,^{20–22} where disruptions in preference for facingness may reduce attention to, or even promote avoidance of, social interaction cues and face stimuli in these clinical populations.

In conclusion, we demonstrate that social interaction history produces lasting, learning-dependent attentional changes. This effect is rapid, automatic, and specific to human conspecifics, challenging the traditional view that higher-order social cognition is inherently slow.⁶¹ By showing that the visual system internalizes abstract social relationships into face representations, we provide a unified theoretical bridge between experience-driven attention and social cognition.

Limitations of the study

This study has two limitations. First, although our stimulus choice was grounded in prior empirical validation that face-to-face dyads serve as a proxy for social interaction, future studies should consider employing more naturalistic forms of social interaction, such as animated dyads or video-based paradigms, to extend our findings to richer social contexts. Second, while we

demonstrate that social interaction history produces learning-dependent changes in attentional capture, it remains unclear how such abstract social information becomes embedded in the visual processing of faces, and whether this reflects perceptual alterations of face representations. Future research using neuroimaging tools could determine whether such social priors are internalized via plasticity within face-selective cortices, marking faces as salient from the earliest stages of visual processing. Addressing this gap is critical for understanding how social experience shapes the architecture of face perception.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Mengyuan Gong (gongmy426@zju.edu.cn).

Materials availability

All task codes have been made publicly available via the Open Science Framework at <https://osf.io/p86t7/>.

Data and code availability

All data and analysis codes have been made publicly available via the Open Science Framework at <https://osf.io/p86t7/>.

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AUTHOR CONTRIBUTIONS

Q. T.: conceptualization, methodology, investigation, formal analysis, visualization, writing—original draft, writing—review and editing; X. Z.: methodology, validation, writing—review and editing; K. J.: writing—conceptualization, review and editing, funding acquisition; M. G.: writing—conceptualization, writing—original draft, writing—review and editing, funding acquisition.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
Experimental data and analysis code	This study	https://osf.io/p86t7/
Software and algorithms		
MATLAB	MathWorks (Version 2020b)	https://www.mathworks.com
Psychtoolbox-3	Brainard ⁶²	https://psychtoolbox.org
JASP	JASP Team (Version 0.17.1)	https://jasp-stats.org

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Participants

To determine the sample size, we ran a pilot experiment with 17 participants using a similar design. We conducted a power analysis using MorePower 6.04.⁶³ Sensitivity analysis indicated that 24 participants would be sufficient to detect an interaction effect ($\eta^2_p = 0.371$) in a two-way (learned configuration $\times$ cue validity) repeated-measures analysis of variance (ANOVA) with a power of 0.95 ($\alpha = 0.05$).

We recruited a total of 144 students from Zhejiang University, assigning them to separate groups of twenty-four for each experiment: Experiments 1a (mean age = 20.50; 15 females, 9 males), 1b (mean age = 22.96; 15 females, 9 males), 2 (mean age = 20.96; 16 females, 8 males), 3a (mean age = 21.92; 11 females, 13 males), 3b (mean age = 21.04; 15 females, 9 males), and 4 (mean age = 20.50; 13 females, 11 males). Informed consent was obtained from each participant in accordance with the Institutional Review Board (2023-007). All participants had normal or corrected-to-normal vision and were right-handed. They were compensated for their participation with ¥ 30 per hour or received course credit.

METHOD DETAILS

Experiment 1

Stimuli and Apparatus

We selected neutral-expression images of eight identities (4 females and 4 males) from the Karolinska Directed Emotional Faces database.⁶⁴ Using Adobe Photoshop CS6, we created three types of dyadic configurations: (1) face-to-face dyad: both individuals oriented toward each other; (2) back-to-back dyad: both individuals oriented away from each other; (3) direct-gaze dyad: both individuals oriented toward the viewer. We created twelve possible dyad combinations, and four pairs were randomly assigned to each participant. Each selected dyad consisted of two faces of the same gender but different identities. All images were converted to gray-scale and equalized for luminance (~ 10 cd/m²). The only modification to the stimuli between Experiments 1a and 1b was that the faces were presented upside down in Experiment 1b.

The experiment was programmed using Psychtoolbox⁶² implemented in MATLAB Version 2020b (The MathWorks, Natick, MA). Stimuli were presented on a 22-inch LCD monitor (resolution: 1920 $\times$ 1080, refresh rate: 100 Hz) against a gray background (~ 5 cd/m²). Participants were seated in a dimly-lit room at a viewing distance of 60 cm, with their head stabilized by a chin rest.

Procedure

Training phase

During the training phase, we used an associative learning task (Figure 1A). Each trial began with a 0.5 s cue display showing two direct gaze faces (same-sex but different identities). Each face subtended $5.5^\circ \times 6.5^\circ$ of visual angle, positioned 3.5° laterally from central fixation, with 1.5° of separation between the faces. Following a jittered delay of 0.1–0.3 s, the target display presented the same face pair in either interacting (face-to-face dyad) or non-interacting (back-to-back dyad) configurations for 0.5 s. Participants judged the configuration as quickly and accurately as possible by pressing the numeric keys (“5” and “6”) within 1.5 s after the target onset. The positions of the faces remained constant between cue and target displays within each trial. For each participant, two face pairs with direct gaze (one female pair and one male pair) were assigned a higher probability of appearing in the interacting configurations (80% of the trials) and a lower probability of appearing in non-interacting configurations (20% of the trials). The other two pairs had the opposite probability assignments. The face pairs were randomly selected for each participant. The training task consisted of 7 blocks (120 trials/block), with each face pair appearing a total of 210 times.

Test phase

During the test phase, we used a cued discrimination task (Figure 1B). Each trial began with a central fixation cross and two lateral boxes (each $1.2^\circ \times 1.2^\circ$) presented for 0.4–0.6 s at 6.4° eccentricity. A cue display then appeared for 0.2 s, showing a direct gaze face dyad (each subtending $3.85^\circ \times 4.55^\circ$ and separated by 1.2°) that had previously been associated with either interacting (face-to-face) or non-interacting (back-to-back) configurations during training. In Experiment 1a, the face dyads were upright, while Experiment 1b presented inverted face dyads to differentiate holistic face processing from low-level facial feature processing. Note that inverted dyads were not used during training in Experiment 1b, as inversion could have impaired the formation of associations between faces and social interactions; this would have made it difficult to determine whether any change in the test effect reflected disrupted learning or disrupted holistic processing. The face dyad was presented adjacent to one of the lateral boxes, with one face positioned on each side of the box. After a blank interval of 0.1–0.2 s, a low-contrast Gabor patch ($1^\circ \times 1^\circ$, tilted $\pm 20^\circ$ from vertical) appeared briefly for 0.05 s, either inside the cued box (valid trials; 50% of the trials) or the uncued box (invalid trials; 50% of the trials). Participants indicated whether the Gabor patch tilted left or right as quickly and accurately as possible by pressing the arrow keys (left-arrow or right-arrow) within 1.5 s after Gabor onset. The task consisted of 3 blocks (96 trials/block), with all trial types randomly interleaved.

Experiment 2

In Experiment 2, we used four arrows that differed in color (red, orange, green and blue) and shape (Figure 3A), adapted from Vestner et al.¹⁶ These arrows were arranged to create three configurations paralleling the face pairs in Experiment 1, including point-to-point (analogous to face-to-face), base-to-base (analogous to back-to-back) and up-down (one pointing upward, the other pointing downward). Arrows were enclosed within circles to reduce the influence of other Gestalt grouping principles (e.g., proximity, good closure). The procedure was largely identical to that in Experiment 1, except for the images used. During the training task, each participant completed 3 blocks (140 trials/block), with each arrow pair presented a total of 210 times. The test task was identical to that in Experiment 1.

Experiment 3

In Experiments 3a and 3b, we used four modified cat faces (adapted from online images; Figure 4A) to create six possible dyad combinations. These stimuli preserved the same spatial configurations as in Experiment 1. The procedure was identical to that in Experiment 1. Each participant was randomly assigned two unique pairs of cat faces.

To assess whether participants had memorized the cat faces during the training task, we included probe trials (20% of all trials) in Experiment 3b. In these probe trials (Figure 5A), an arrow appeared after participants responded to the target display (face-to-face or back-to-back cat faces), pointing either left or right. The arrow was followed by a probe display containing four cat faces, and participants had to select the cat faces shown at the probed location from the preceding target display. Feedback was then provided to indicate whether their response was correct. The remaining 80% of trials were identical to the training task in Experiment 3a.

Experiment 4

The stimuli and procedure were identical to those in Experiment 1, except that during the test task, a single face ($3.85^\circ \times 4.55^\circ$) with direct gaze was presented in the cue display.

QUANTIFICATION AND STATISTICAL ANALYSIS

Behavioral performance

For each participant and each experimental condition, we separately calculated accuracy and reaction time (RT) for both the training and test tasks. RT analyses included only correct responses within the predefined response window (0.2–1.5 s). We excluded outlier trials where RTs exceeded three standard deviations from each participant's mean.

Bayesian analysis

To evaluate the strength of evidence for the lack of an interaction effect, we conducted parallel Bayesian analyses²⁸ using standard priors in JASP Version 0.17.1.⁶⁵ Specifically, we compared a null model containing two main effects and subject-related effects to an interaction model that included all terms of the null model plus the interaction.